

Mohammad Fathi

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RESEARCH INTERESTS

- Machine Learning and Deep Learning
- Computational Neuroscience
- Graph Neural Networks
- Brain–Computer Interfaces and Neuroimaging (EEG, fNIRS, fMRI)

EDUCATION

Kharazmi University

Sep. 2021 – Sep. 2024

M.S. in Data Mining (Computer Engineering) – GPA: 3.74/4.0 (17.25/20), ranked 2nd in cohort Tehran, Iran

– Thesis: *Diagnosis of Mild Cognitive Impairment (MCI) Using Graph Neural Networks Based on fNIRS Data* (grade: 20/20)

– Advisor: Dr. Mir Mohsen Pedram

Shahed University

Sep. 2016 – Jul. 2021

B.S. in Computer Science – Full tuition waiver scholarship

Tehran, Iran

PUBLICATIONS

Accepted

- **M. Fathi**, M. Faramarzi, M. Gholami, N. Deravi. “Calibrated Ensemble Deep Learning for Four-Class Lung CT Image Classification and Uncertainty-Based Referral.” *Digital Health* (SAGE).

Submitted

- **M. Fathi**, M. Faramarzi, M. Gholami, M.M. Pedram. “FAT-Net: A Frequency-Aware Temporal Graph Network Recovering Clinically Valid Dementia Signatures Where Aggregation May Bias Gradient-Based XAI.” *Neuroinformatics* (Springer).

RESEARCH AND TEACHING EXPERIENCE

Research Assistant

Sep. 2022 – Nov. 2024

Kharazmi Cognitive and Brain Science Lab, Kharazmi University

Tehran, Iran

- Acquired and preprocessed EEG and fNIRS data for brain–computer interface and neuroimaging studies; developed preprocessing guidelines for the lab’s fNIRS pipeline.
- Trained master’s students to operate the fNIRS device and acquire data under the supervision of Dr. Pedram.

Teaching Assistant, Machine Learning

Oct. 2022 – Feb. 2023

Kharazmi University

Tehran, Iran

- Delivered classroom sessions, designed and graded homework assignments, and held individual consultations with students.

INDUSTRY EXPERIENCE

Data Scientist (RAG Systems & Fraud Detection)
Bahar Samaneh

Jan. 2026 – Present

- Designed and implemented Retrieval-Augmented Generation (RAG) systems using embedding models, vector databases, and semantic search over large-scale text data.
- Built an educational AI chatbot that answers questions from school textbooks, plus an LLM-based exam and quiz generator.
- Developed machine learning pipelines for fraud and anomaly detection.

Data Scientist
Atitarh System Saman

Oct. 2024 – Aug. 2025

- Built anomaly detection and time-series forecasting models (including deep learning) for leakage detection and predictive maintenance in water distribution networks.
- Cleaned and preprocessed large-scale sensor data; integrated models into real-world water management systems with engineers and stakeholders.

INVITED TALKS

AI in Action: From Smart Vehicles and Computer Vision to Neuroscience
Research Week, Kharazmi University

Dec. 2024
Tehran, Iran

HONORS AND AWARDS

- Ranked 2nd among master’s students in Data Mining, Kharazmi University
- Full Tuition Waiver Scholarship, Shahed University (2016–2021)

TECHNICAL SKILLS

Languages:	Python, MATLAB, SQL, C++
ML Frameworks:	PyTorch, PyTorch Geometric (incl. PyG Temporal), TensorFlow, Keras, scikit-learn, CUDA
Neuroimaging:	MNE-Python, EEGLAB
Data Tools:	NumPy, pandas, Matplotlib, Seaborn, Tableau
Tools:	Git/GitHub, Jupyter, VS Code

LANGUAGES AND TEST SCORES

English:	TOEFL iBT 105/120 (Sep. 2025) – Reading 30, Listening 30, Speaking 21, Writing 24
Persian:	Native

REFERENCES

Dr. Mir Mohsen Pedram Associate Professor, Kharazmi University pedram@khu.ac.ir	Dr. Niloofar Deravi, MD School of Medicine, Shahid Beheshti University of Medical Sciences niloofarderavi@sbmu.ac.ir
Dr. Meghdad Mirabi Assistant Professor, TU Darmstadt meghdad.mirabi@cs.tu-darmstadt.de	Dr. Manoochehr Kelarestaghi Assistant Professor, Kharazmi University Kelarestaghi@khu.ac.ir